

Introduction

This document provides a detailed guide on accessing, extracting, and interpreting patient information stored within an Electronic Health Information (EHI) file. This file is generated by the CloudMD365 system as part of its patient record export functionality. Specifically, the guide will cover:

- **File Format and Structure:** An overview of the EHI file format, layout, and contents, which may include patient demographics, medical history, medications, treatment plans, and other relevant health records.
- **Extraction Process:** Step-by-step instructions on how to download or retrieve the EHI file from CloudMD365, along with any tools or software needed for extraction.
- **Understanding Patient Information:** Guidance on reading and understanding each section of the EHI file, including tips for locating specific types of patient data and interpreting any abbreviations or codes used within the document.

This document is intended for healthcare professionals, IT staff, or anyone responsible for managing patient records who requires a clear understanding of the EHI file generated by CloudMD365.

File Format and Structure

The Electronic Health Information (EHI) file exported from CloudMD365 is typically structured in a standardized format, such as JSON or XML. This ensures compatibility with various health information systems and ease of use for data analysis and integration. Below are the key components and organization of the EHI file.

File Format

The EHI file is generally saved in either **JSON** (JavaScript Object Notation), which is a human-readable format that facilitates data exchange between systems.

A JSON file may look like this:

Sample JSON Content
{

```
{
  "patient": {
    "id": "123456",
    "name": {
      "first": "John",
      "last": "Doe"
    },
    "dob": "1990-01-01",
    "gender": "male"
  }
}
```

Key Sections

- **Patient Demographics:** Contains essential identifying details for the patient, including name, date of birth, gender, address, and contact information. This section ensures accurate patient identification and is critical for interoperability with other systems.
- **Health Summary:** Provides a high-level overview of the patient's health status, including primary diagnoses, chronic conditions, past surgeries, and relevant medical history. This summary allows healthcare providers to quickly assess the patient's overall health context.
- **Medications:** Lists all prescribed and over-the-counter medications, including drug names, dosages, and administration schedules. Information on medication history, start and end dates, and prescribing provider is also included.
- **Allergies:** Documents known allergies, including medications, environmental factors, and food allergens, as well as the severity of reactions. This section is essential for avoiding adverse drug interactions and ensuring patient safety.
- **Encounters:** Records details of past medical visits, including encounter dates, visit types (e.g., in-person, telemedicine), reasons for the visit, diagnoses made, treatments provided, and follow-up recommendations. This section helps in tracking the patient's medical history and continuity of care.
- **RPM Data (Remote Patient Monitoring):** Contains data collected from remote patient monitoring devices, such as blood pressure readings, glucose levels, heart rate, and other relevant metrics. This section is valuable for tracking patients' health in real time, especially those with chronic conditions requiring frequent monitoring.

Data Standards

The EHI file adheres to standards like HL7 FHIR to ensure interoperability, enabling the data to be integrated smoothly across different healthcare platforms.

Extraction Process

To successfully extract patient information from the EHI file generated by CloudMD365, follow the steps outlined below:

1. Log into CloudMD365
 - Access the CloudMD365 platform using your authorized credentials. Ensure that you have the necessary permissions to access patient data.
2. Navigate to the Export Functionality
 - Once logged in, from the menu bar, locate the Admin > Export Patient Health Data. Then, select the patients to export data.
3. Specify Export Options
 - Set the parameters such as data range for encounters to include in the export.
4. Initiate the Export
 - Click the Export button to generate the EHI file. This may take a moment depending on the amount of data being processed.
5. Download the File:
 - Once the export is complete, a download link or prompt will appear. Click to download the EHI file to your local machine or designated storage location.

Store the downloaded EHI file securely in compliance with data protection regulations (e.g., HIPAA) to maintain patient confidentiality and data integrity.

Understanding Patient Information

This section provides instructions on how to interpret the data within each part of the Electronic Health Information (EHI) file generated by CloudMD365. Understanding the structure and content of each section is essential for healthcare providers and administrative staff to accurately review patient information.

Patient Demographics

This section contains basic identifying information about the patient, such as their name, date of birth, gender, address, and contact details. It serves as a foundational reference to confirm the patient's identity and ensure that their medical records are accurately attributed.

Below table is the list of attributes with specified data type and short description.

PATIENT DEMOGRAPHICS		
Field Name	Data Type	Short Description
patient_id	GUID	Unique identifier for the patient in the system
first_name	String	Patient's first name
last_name	String	Patient's last name
date_of_birth	String	Patient's date of birth in YYYY-MM-DD format
gender	String	Patient's gender
address	Object	Object containing address details for the patient
email	String	Patient's email address

ADDRESS		
Field Name	Data Type	Short Description
street	String	Street address of the patient
city	String	City of the patient's residence
state	String	State of the patient's residence
zip_code	String	Postal or ZIP code of the patient's address
home_phone	String	Patient's home phone number
work_phone	String	Patient's work phone number

Health Summary

The Health Summary section provides a high-level overview of the patient's current and past health conditions. It includes primary diagnoses (especially chronic or long-term conditions), a history of surgeries, and relevant family medical history. This section gives healthcare

providers a quick snapshot of the patient's health background, which aids in treatment planning.

HEALTH SUMMARY		
Field Name	Data Type	Short Description
chronic_conditions	Array of Problem	List of Patient's Current Chronic Conditions
diagnosis	Array of Problem	List of Patient's Diagnosis
family_history	Array of String	List of family history

PROBLEM		
Field Name	Data Type	Short Description
problem	String	Problem Name
icd_10	String	Corresponding ICD-10

Medications

This section lists all current medications prescribed to the patient, along with details like dosage, frequency, and the prescribing provider's information. By reviewing this information, providers can monitor medication regimens, avoid potential interactions, and ensure adherence to treatment plans.

MEDICATIONS		
Field Name	Data Type	Short Description
medication	Array of Object	Name of the Medication
dosage	String	Dosage Information for the medication
frequency	String	Frequency of administration

start_date	DateTime	Date the medication was started
end_date	DateTime	Date the medication was ended

Allergies

The Allergies section records any known allergies the patient has, detailing the allergen, the type and severity of the reaction, and the last time the reaction occurred. This information is crucial to prevent adverse reactions to medications or treatments and is essential for safe clinical decision-making.

ALLERGY		
Field Name	Data Type	Short Description
allergy	String	Name of Allergy
type	String	Type of Allergy (Ex. Drug)
reaction	String	Allergy Reactions

Encounters

The Encounters section documents each of the patient's healthcare visits or interactions, capturing the date, visit type, reason for the visit, diagnosis (with ICD-10 codes), treatment provided, and care plan details. It also includes the provider's notes. This section gives a timeline of the patient's medical history and allows for continuity of care, helping providers understand the patient's health journey and progress.

ENCOUNTER		
Field Name	Data Type	Short Description
encounter_date	DateTime	The date of the encounter, formatted as YYYY-MM-DD.
chief_complaint	String	The primary reason for the encounter.

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diagnosis	Array of Problem	Contains details of the diagnosis, including the condition and ICD-10 code.
treatment	String	Summary of the treatment provided during the encounter.
care_plan_goals	Array of String	List of specific health goals set for the patient.
care_plan_actions	Array of String	List of actions or recommendations for the patient.
care_plan_followup	String	Instructions for follow-up or reassessment.
provider_id	GUID	ID of the provider
provider_name	String	Name of the provider responsible for the encounter
notes	String	Additional comments or observations from the provider

RPM Data (Remote Patient Monitoring)

This section includes data collected from remote monitoring devices, such as blood pressure, glucose levels, and heart rate readings. These real-time metrics are valuable for tracking chronic conditions and providing proactive care, as they allow providers to monitor health trends outside of the clinic setting.

RPM DATA		
Field Name	Data Type	Short Description
recorded_time	DateTime	Recorded time of the RPM Data
data_type	String	Type of recorded data (Ex. blood_pressure)
attribute_type	String	Type of attribute for the specific data type (ex. systolic)
value	Float	Value of the recorded data
unit	String	Unit of the recorded data